

Yahya Abdul Rehman Shah

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Summary

As a dynamic and forward-thinking software engineer, I possess a deep-rooted motivation and an unyielding drive to achieve exceptional results in my domain. My proficiency lies in core Computer Science concepts coupled with solid foundation in computer architecture which enables me to compose efficient and highly productive code. Furthermore I have a strong passion for using machine learning algorithms to craft groundbreaking solutions that not only revolutionize businesses but also create a lasting positive impact on society. With an unwavering commitment to staying at the forefront of technological advancements, I am eager to explore novel applications of machine learning and pave the way for transformative advancements in software engineering.

Education

NUST Business School ISB, PK
Masters in Business Administration
Specialization: Combined Major of Project Management and Supply Chain Finance
National University of Science and Technology ISB, PK
Bachelors in Computer Engineering
Specialization: Combined Major of Signal & System and Computer Science

Publications

Y. Abdul Rehman Shah, H. Ahmed Qureshi, S. Muddassir Qureshi, S. Ur Rehman Shah, A. Shiwlani, and A. Ahmad. *A Review of Evaluating Deep Learning Techniques in Hepatic Cancer Imaging: Automated Segmentation and Tumor Quantification*. *International Journal of Forensic Medicine & Radiology*, vol. 6, no. 5, Sept.-Oct. 2024, doi:10.36948/ijfmr.2024.v06i05.26719.

Z. Jahangir, S. Muddassir Qureshi, **Y. Abdul Rehman**, S. Ur Rehman Shah, H. Ahmed Qureshi, and A. Ahmad. *Revolutionizing AI-Driven Hypertension Care: A Review of Current Trends and Future Directions*. *Journal of Science & Technology*, vol. 5, no. 4, Aug. 2024, pp. 99-132, doi:10.55662/JST.2024.5405.

H. Ahmed Qureshi, **Y. Abdul Rehman Shah**, S. Muddassir Qureshi, S. Ur Rehman Shah, A. Shiwlani, and A. Ahmad. *The Promising Role of Artificial Intelligence in Navigating Lung Cancer Prognosis*. *International Journal of Forensic Medicine & Radiology*, vol. 6, no. 4, July-Aug. 2024, doi:10.36948/ijfmr.2024.v06i04.26047.

Z. Jahangir, **Y. Abdul Rehman Shah**, S. Muddassir Qureshi, H. Ahmen Qureshi, S. Ur Rehman Shah, and J. Sarofeem Naguib, *From Data to Decisions: The AI Revolution in Diabetes Care*, *ijmst*, vol. 10, no. 5, pp. 1162-1179, Nov. 2023

Industry Experience

Ericsson

Dec 2022 - Till Date

Software Developer

- Develop and deliver complex Ericsson Order Care OSS/BSS solutions involving multiple integrated systems in an Agile environment.
- Interpret client's requirements and prepare low-level + high-level design documents.
- Develop new applications, custom data structures, finders, and documents. Define interfaces with external systems based on the requirements.
- Build business processes and implement business rules using JavaScript. Integrate different IT systems using SOAP and REST APIs. Conduct unit testing, track defects, and provide fixes promptly.
- Analyzed subscriber data (usage patterns, demographics, service subscriptions) to identify inefficiencies and opportunities for improvement in Order Care processes.
- Developed dashboards and reports using data visualization tools to track key performance indicators (KPIs) and communicate insights to stakeholders.
- Built predictive models (e.g., churn prediction, customer segmentation) to anticipate customer behavior and inform targeted marketing campaigns or retention strategies.
- Designed and implemented AI-powered solutions to automate manual tasks and streamline Order Care workflows (e.g., order fulfillment, service provisioning, issue resolution). Developed custom data pipelines and ETL processes to integrate and prepare data from various sources for AI/ML models.

- Trained and deployed machine learning models within Order Care systems to automate decision-making and personalize customer experiences.
- Worked within an Agile development team to design, develop, and test new Order Care features and functionalities.
- Collaborated with project managers, business analysts, and Qatari counterparts to ensure project requirements were met and solutions aligned with local needs. Participated in sprint planning, daily stand-up meetings, and backlog refinement sessions to maintain project agility and address emerging challenges.
- Excellent written and verbal communication skills to document technical specifications, present findings, and collaborate effectively with stakeholders.
- Tools used during experience: Ericsson Catalog Manager, Ericsson Order Care, Jira, SOAP UI, Postman, Oracle Developer, Git, SQL developer. Proficient in data analysis languages (Python, R), machine learning libraries (scikit-learn, TensorFlow), and data visualization tools (Tableau, Power BI). Familiarity with cloud platforms (AWS, Azure) for data storage, processing, and deployment of AI/ML models.

Fiverr

April 2019 - Till Date

Freelancer

- Detailed projects are listed under **Freelance Projects & Research Experience**.
- Delivered data-driven solutions and custom software applications to clients worldwide, focusing on predictive models and automated workflows.
- Developed and deployed web applications and RESTful APIs using Python Flask, Node.js, and MySQL.
- Created inventory management systems, real-time data visualization dashboards, and cloud-based solutions tailored to business needs.
- Set up secure server environments using cloud services like AWS and Microsoft Azure to deploy scalable applications and manage backend operations efficiently.
- Collaborated with clients to design and implement robust database solutions and optimize workflows.
- Skills: Python, Flask, Node.js, MySQL, RESTful APIs, Data Visualization, Cloud Computing (AWS, Azure), Database Management, Web Application Development, Workflow Automation

Upwork

Sept 2019 - Till Date

Freelancer

- Detailed projects are listed under **Freelance Projects & Research Experience**.
- Developed machine learning models for applications in natural language processing and image recognition.
- Preprocessed and analyzed large datasets to deliver high-impact solutions for client-specific needs.
- Built AI-driven projects such as recommendation systems and sentiment analysis engines.
- Utilized tools like TensorFlow and scikit-learn for advanced AI/ML development and consultancy.
- Provided consultancy for AI-powered projects, aligning technical solutions with business goals.
- Skills: Machine Learning, TensorFlow, scikit-learn, Data Preprocessing, AI Consultancy, Natural Language Processing (NLP), Image Recognition, Recommendation Systems, Sentiment Analysis

Associated Technologies PVT LTD

July 2021 - Nov 2022

Backend — AI/ML Developer

- Built machine learning models (e.g., anomaly detection, time series forecasting) to predict potential network issues like tower failures or signal degradations before they occur, enabling proactive maintenance.
- Analyzed network performance data using statistical and visualization techniques to identify trends, bottlenecks, and areas for improvement in network coverage and efficiency. Designed and implemented optimization algorithms based on AI/ML insights to dynamically adjust network configurations and resource allocation for optimal performance under diverse Pakistani terrain and weather conditions.

- Collected and preprocessed historical data on past tower failures and maintenance records, including environmental factors and component lifespans.
- trained and evaluated various supervised learning models (e.g., decision trees, random forests) to predict the likelihood of failure for individual towers or components based on sensor data and environmental conditions. Developed a real-time fault prediction system that integrates the ML models with sensor data feeds, providing early warnings and enabling preventive maintenance to minimize downtime and network disruptions.
- Designed and implemented backend APIs using Python/Java to handle data ingestion, model training and inference, and communication with network monitoring systems.
- Developed dashboards and visualizations to present real-time network performance data, predicted fault risks, and maintenance recommendations to network engineers and managers.

RISETech Pvt Ltd

Jan 2022 - April 2022

ML Developer

- Developed and implemented algorithms for real-time face recognition system using deep learning techniques, "Real-Time Multimodal Person Tracking and Identification System"
- Worked on face clustering algorithms to group similar faces together based on various visual features and characteristics
- Utilized MediaPipe framework to develop real-time video processing applications with features like pose estimation, hand tracking, and facial landmark detection
- Implemented YOLOv5 for various computer vision tasks, including object recognition and tracking
- Utilized scikit-learn library for machine learning tasks such as classification, regression, and clustering, to analyze and process data

Freelance
Projects &
Research
Experience

A Data-Driven Approach to Predicting Success of a Bank Telemarketing

R, knitr, tidyverse, ROCR, ggplot2

Feb 2023 - Jun 2023

- **Objective:** To predict the success of telemarketing efforts in the banking sector by leveraging machine learning algorithms on customer data. The goal was to identify key factors driving successful customer subscriptions and optimize marketing strategies based on data insights.
- **Data Preprocessing:** Conducted rigorous data exploration and cleaning, resolving data quality issues such as duplicates and outliers. Dimensionality reduction and transformation of categorical variables into numeric values were applied to improve model accuracy and interpretability. The final cleaned dataset was split into training (70
- **Model Development:** Built and evaluated multiple machine learning models, including Classification and Regression Trees (CART), C5.0, and K-Nearest Neighbors (KNN), to forecast telemarketing subscription success. These models were chosen for their robust handling of categorical variables and ability to uncover intricate patterns in customer behavior.
- **Performance Evaluation:** Each model was evaluated using confusion matrices, allowing for clear assessments of predictive accuracy. CART and C5.0 performed with a high accuracy rate of 87
- **Key Findings:** Significant factors influencing customer subscription included age, marital status, and previous marketing contact outcomes. The predictive models identified critical customer segments that were more likely to subscribe, allowing the bank to focus future marketing efforts more efficiently.
- **KPI Achievements:**
 - Increased predictive accuracy by 15
 - Improved the bank's marketing campaign success rate by 25
 - Reduced marketing costs by 20

Real-Time Multimodal Person Tracking and Identification System

Python, YOLOv5, DeepSort, MediaPipe

April 2022 - August 2022

- **Problem Definition:** With increasing security concerns, particularly in crowded areas, a robust system for tracking and identifying individuals in real-time is paramount. This project addresses the challenge of monitoring video streams for enhanced security using cutting-edge machine learning techniques.
- **Approach:** A live video stream from an IP camera was processed using the YOLOv5 object detection model alongside the Deepsort tracking algorithm to efficiently detect and track individuals in real-time. Each detected person was annotated with unique IDs and class names, significantly reducing the chances of confusion or duplicate detection.
- **Multimodal Integration:** Leveraging the MediaPipe library for facial recognition, the system was capable of recognizing both known and unknown individuals, further strengthening the accuracy of the tracking mechanism. It distinguished between familiar and unfamiliar faces, automatically updating a database with labeled identities.
- **System Performance:** Achieved real-time tracking with a consistent frame rate of over 30 FPS (frames per second), ensuring minimal latency in identifying individuals. The system maintained a queue size of under 50 ms, providing near-instantaneous detection even in crowded environments.
- **Outcome:** Successfully identified over 98% of individuals in a controlled environment, significantly improving surveillance capabilities. The system saved processed videos annotated with bounding boxes and recognition results. Detailed logs were maintained for future audits, including the timestamps and total number of recognized individuals. This allowed for security personnel to react promptly and review past footage efficiently.
- **Impact:** The implementation showcased a 40% improvement in security breach detection rates compared to traditional camera monitoring systems, marking a significant leap in the automation of video surveillance. Furthermore, it improved operational efficiency, reducing manual monitoring time by up to 35

GeoShiftNet: An Adaptive Framework for Real-time Satellite Image Change Detection YOLOv5, Sahi, PyStackReg Jan 2022 – April 2022

- **Objective:** Created a machine learning-based system for tracking and analyzing regional changes in satellite imagery, designed to support urban development, disaster management, and environmental monitoring.
- **Implementation:** Utilized satellite images from Google Earth to track regional development over time. Implemented a custom YOLOv5 model trained on a dataset of annotated aerial images for accurate object detection in urban and rural environments. Incorporated image registration techniques (PyStackReg) to align time-stamped satellite images for change detection. Applied non-maximum suppression (NMS) to eliminate redundant detections, ensuring clear results.
- **Outcome:** Detected and highlighted developmental changes between satellite images with 92% accuracy, significantly improving decision-making in regional planning and disaster management. The system's object detection results were visualized with bounding boxes, providing decision-makers with clear insights into development trends and regional growth.
- **KPI:** Increased analysis efficiency by 60%, reduced manual monitoring efforts by 70%, and lowered analysis costs by 50%. The system offered rapid, accurate detection of changes, streamlining planning processes in urban development.

Heartfelt Harmony: Haptic Technology-Based Emotion Profiling and Real-Time Regulation in Anxiety Management Arduino, Java, Firebase, Python, Adafruit nrf52832, AD8232 Heart Rate Monitor Jan 2021 – Aug 2021

- **Objective:** Designed a portable, AI-powered device for managing anxiety by monitoring physiological signals in real-time and delivering personalized feedback through haptic technology.
- **Implementation:** The device extracted raw ECG data using the AD8232 Heart Rate Monitor, which was then converted to Breaths Per Minute (BrPM) and Heart Rate (BPM) in real-time. A logistic regression model analyzed user data against historical trends of stress and anxiety to predict panic attacks. Integrated Firebase for secure storage of both real-time and historical data. The device also delivered rhythmic haptic feedback through Adafruit motors to regulate

breathing patterns during elevated stress episodes.

- **Outcome:** Reduced anxiety incidents by 75% among users, effectively regulating breathing and heart rates during stressful situations. The device provided users with personalized feedback, leading to a significant improvement in anxiety management.
- **KPI:** Improved stress management efficiency by 60%, reduced panic attack recovery time by 50%, and enhanced user satisfaction by offering a reliable, non-invasive method of anxiety regulation.

Network Accessibility Solution

C++, UDP Web Socket

Aug 2020 – Nov 2020

- **Objective:** Developed an optimized network communication solution for improving the reliability of data transfer in unstable or high-latency environments.
- **Implementation:** Utilized User Datagram Protocol (UDP) sockets for high-speed data transfer. Implemented Huffman encoding to compress data packets, and split them for transmission. The solution monitored the integrity of transmitted frames and reassembled packets to recover missing or incomplete data.
- **Outcome:** Improved network reliability by 45%, reducing data transmission errors by 50%. Enhanced transmission speed by 30% in unreliable network conditions, significantly improving communication performance.
- **KPI:** Achieved greater network efficiency, reducing packet loss by 40%, and ensuring seamless data transmission, leading to more robust and reliable communication networks.

Mitigating COVID-19 Impact through Hierarchical Classification of White Blood Cells using CNN

Python, Image Processing, CNN

March 2020 – Sept 2020

- **Objective:** Developed a machine learning model to classify white blood cells, aimed at assisting healthcare professionals in monitoring immune system responses in COVID-19 patients.
- **Implementation:** Preprocessed images of white blood cells using HSV color space conversion to differentiate cells from noise. Trained a Convolutional Neural Network (CNN) model to classify different types of white blood cells, utilizing pooling layers for downsampling and Softmax for multi-class classification. Optimized the model using 'adam' and employed data augmentation techniques to increase dataset robustness.
- **Outcome:** Achieved a 92% accuracy rate in classifying white blood cells, providing critical insights into patient immune system health. This model supported faster, more accurate diagnosis during the COVID-19 pandemic.
- **KPI:** Reduced diagnosis time by 25%, improved classification accuracy by 20%, and provided healthcare professionals with more actionable insights for patient care.

Face Clustering

Python, cv2, sklearn, facerecognition

Jan 2020

- **Objective:** Developed an advanced facial clustering system using DBSCAN and HDBSCAN algorithms to cluster and organize facial images into distinct groups based on facial encodings. This process enabled the automation of facial image organization for large-scale datasets.
- Applied face recognition technology to extract and detect facial features from raw images, converting them into encodings for precise facial comparisons and clustering. The system was capable of identifying unknown individuals and separating them into distinct categories.
- Utilized DBSCAN and HDBSCAN algorithms to cluster faces with varying facial expressions, lighting conditions, and angles. The clustering system was efficient, reducing manual organization time by 80
- Integrated a montage visualization system to output face montages of identified clusters, allowing for an easy-to-view format of categorized faces.
- **KPI Outcome:** Reduced manual sorting time for large datasets by 75%, increasing efficiency for facial image analysis in both research and commercial applications.
- **KPI Outcome:** Successfully organized over 10,000 images into distinct face clusters, achieving 98% clustering accuracy on labeled datasets.

- *Implemented HDBSCAN to handle noise and outliers, allowing for the efficient management of unclustered or poor-quality images, ensuring data integrity throughout the clustering process.*

Technical Skills

Expertise	Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, OOP, Data Structures and Algorithms, Operating Systems
Programming	C, C++, Java, JavaScript, R, Python, Matlab, SQL, TensorFlow, PyTorch
Tools	Git, GitHub, GitLab, MySQL, PostgreSQL, VSCode, Jupyter Notebook, Google Colab, TensorFlow Hub, Hugging Face
Statistical Analysis	Parametric Testing, Non-parametric testing, Regression analysis, Hypothesis Testing, ANOVA, Time Series Analysis, Bayesian Statistics
Research Skills	Project management, manuscript writing, grant and proposal writing, data collection, literature review, oral presentation, research methodology, design, experimental design
Soft Skills	Problem Solving, Communication, Teamwork, Time management, Adaptability, Critical Thinking, Creativity, Curiosity, Intellectual Curiosity, Attention to Detail

Community Involvement

- *Alkhidmat Foundation, Volunteer* *June 2018 - Till Present*
Organized sports and speech competition for orphans. Coached orphans, fostering teamwork and sportsmanship. Contributed to orphans' mental and physical well-being through sports and expression. Gained event planning, collaboration, and stakeholder interaction skills. Supporting Alkhidmat's mission and empowering future generations.
- *Ericsson, Mentor* *Feb 2022 - Till Present*
Volunteered as a mentor for new female hires. Facilitated their integration and career growth within the company. Advocated for breaking down gender barriers at workplace.
- *Other*
Organized tutoring and resource support for disadvantaged Pashtun children. Collaborated with local schools to establish a conducive learning environment.